SMART INDIA HACKATHON 2024

- Problem Statement ID - SIH1739
- Problem Statement Title - Building Integrated Photo-voltaic (BIPV) potential assessment and visualisation using LOD-1 3D City Model
- Theme - Renewable / Sustainable Energy
- PS Category- Software
- Team Name - Operation_VIJAY

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3D BIPV POTENTIAL WITH LOD-1 MODELS

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Scope of the Problem

- Building Integrated Photovoltaic (**BIPV**) and **Rooftop PV potential** need assessment for urban areas using satellite data.
- Lack of existing **public domain tools** to simulate solar energy potential on vertical and rooftop surfaces of buildings.
- Need for scalable, city-wide renewable energy assessment tools for policy-makers, solar solution providers, and architects.

Solution

- Interactive application utilizing **LOD-1 3D city models** derived from satellite data (Cartosat-2/3 and Cartosat-1).
- Simulates **solar energy radiation** on building surfaces based on user-provided date and Global Horizontal Irradiation (GHI) values.
- Dynamic 3D visualization that estimates and displays BIPV and rooftop PV energy potential.
- Accounts for **shading effects** from nearby buildings and structures, providing accurate estimations.

City-Wide Scale

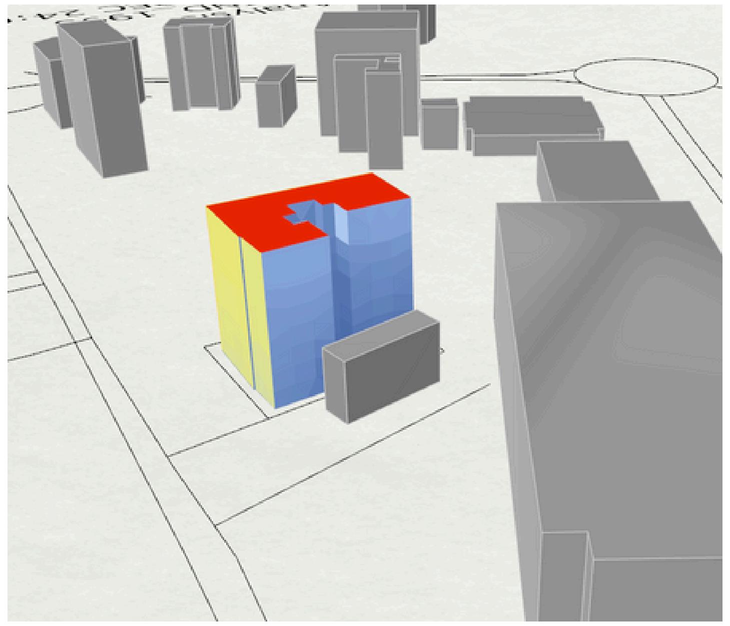
Simulates Real Conditions

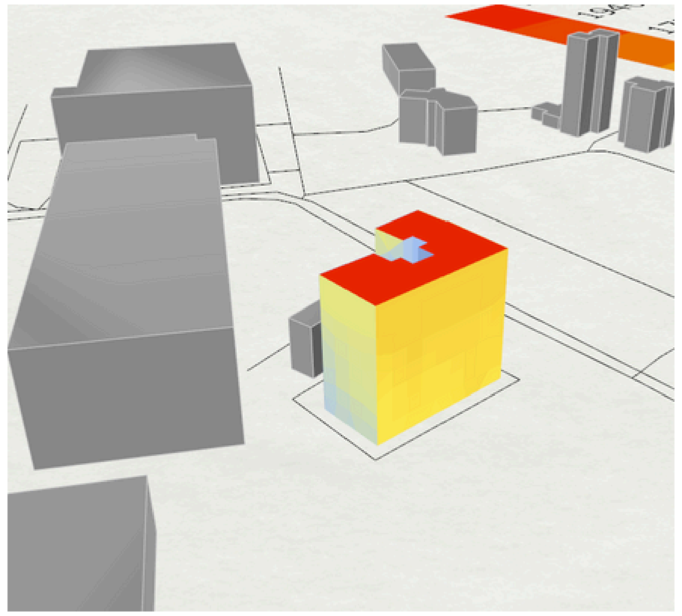
User-Friendly

Accessible to All Stakeholders

Innovation/Uniqueness

- Integration of LOD-1 models for city-wide **BIPV potential assessment**, filling the **gap in current city-level renewable** energy tools.
- Shadow simulation to estimate solar energy on vertical surfaces, expanding the traditional focus on rooftop PV.
- Dynamic, real-time **calculations based on sun position**, allowing users to explore energy potential for any given day of the year.
- Scalable and lightweight solution, unlike detailed architectural tools that require high-level of software expertise.



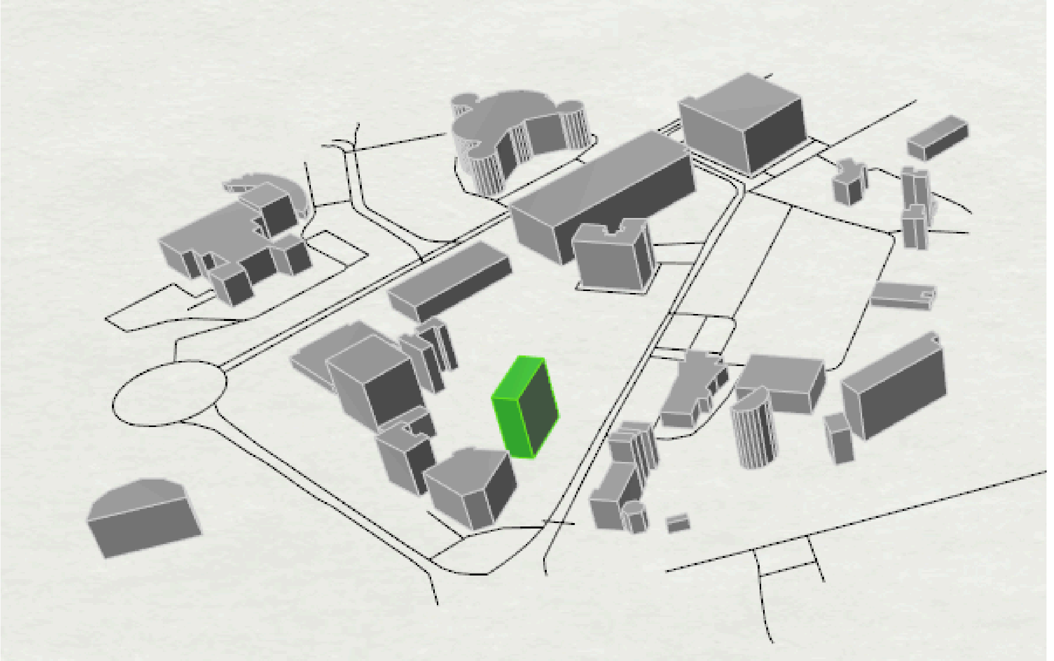


3D visualization with energy potential estimation

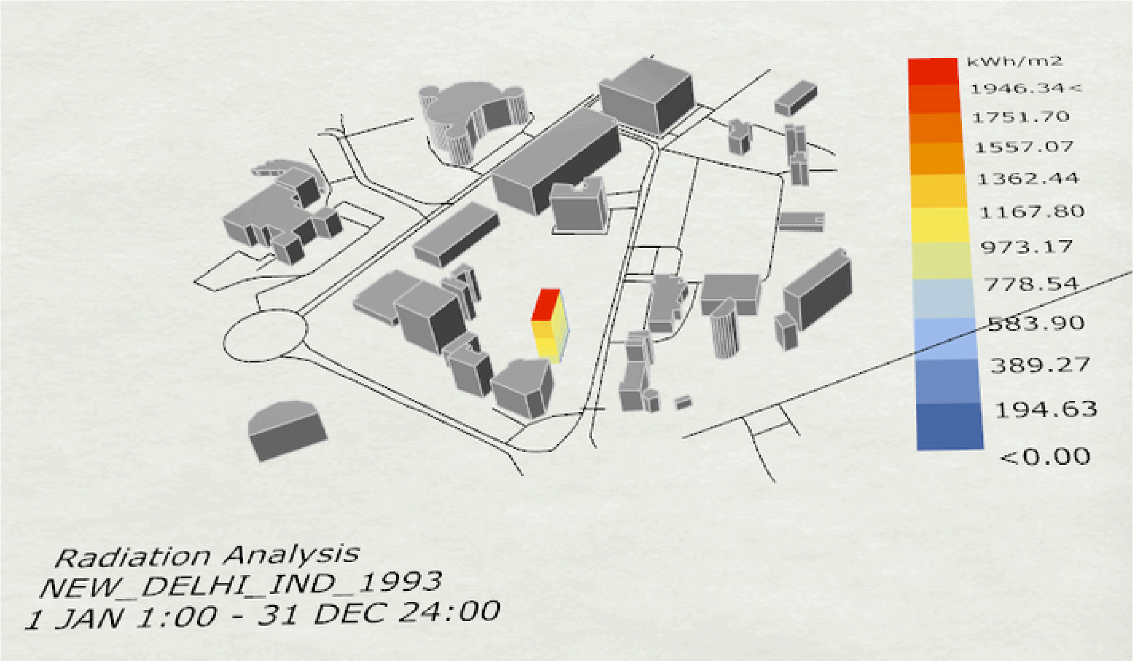
TECHNICAL APPROACH

Site Selection

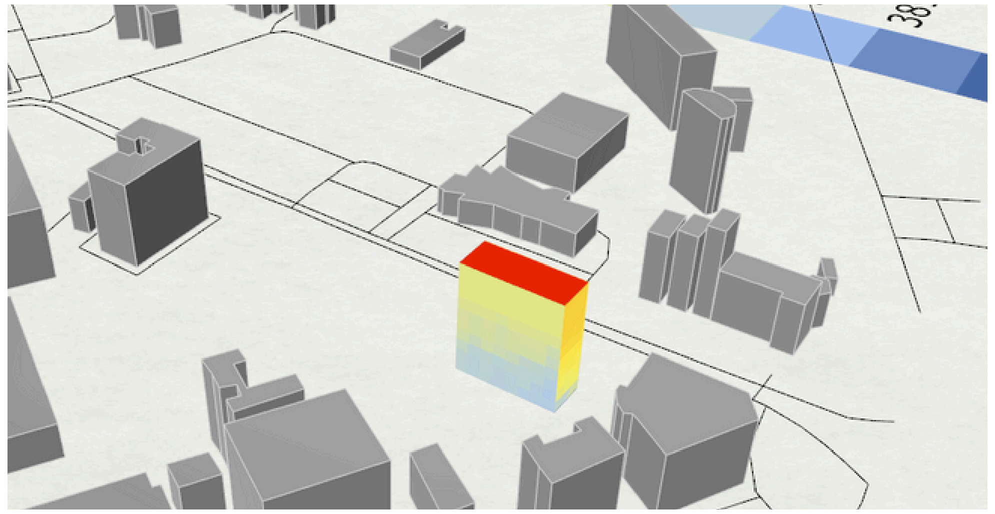
Building footprint tracing



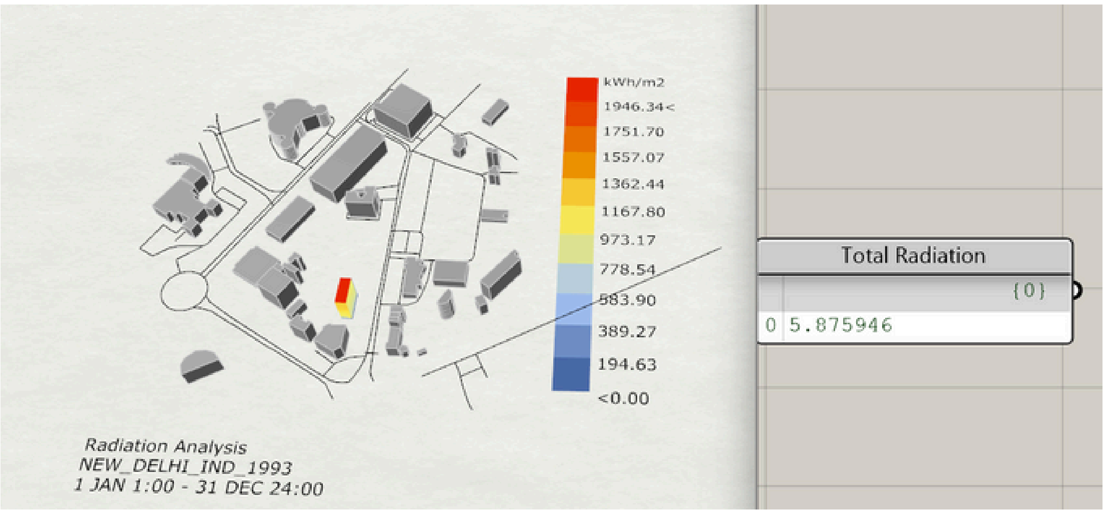
LOD 1 model creation



Shadow simulation and Solar energy calculation



Energy Potential on building surface



Total Radiation on building facade

PV Potential Calculation

Solar Energy Output = S × IRR × PR × Efficiency

S = Surface Area, IRR = Incident Radiation

PR = Performance Ratio, Efficiency = Efficiency of Solar Modules

Source

Development

CSS HTML node JS mongoDB React

Simulation and Energy assessment

grasshopper

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FEASIBILITY AND VIABILITY

Feasibility

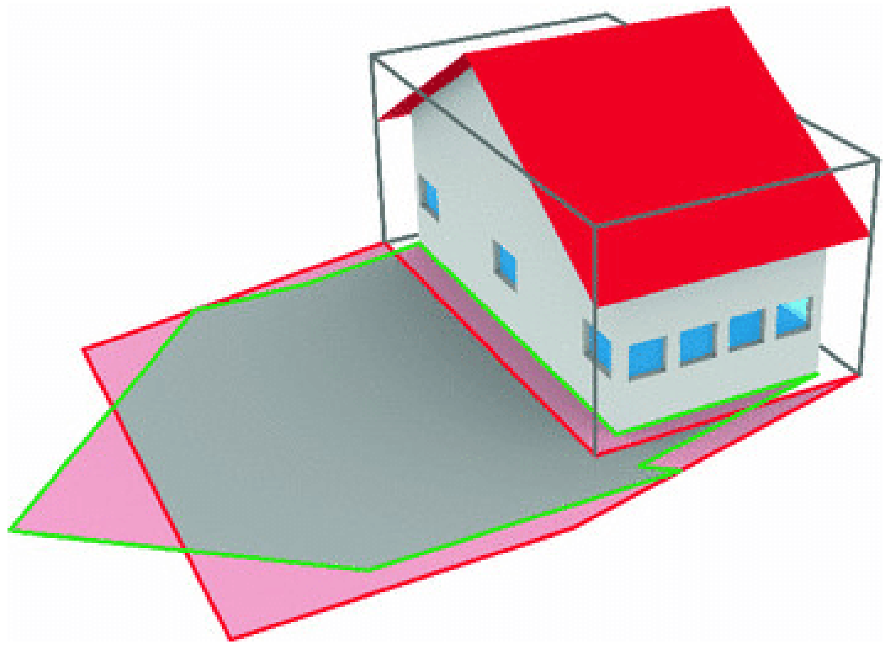
- **Available Satellite Data**: Cartosat-1, 2/3 data enables generating LOD-1 3D city models
- **Proven Simulation Tools**: Existing shading and solar radiation models can be adapted
- **Generalizing**: Enable energy potential assessments for any building without being dependent on specific requirements or conditions for implementation.
- **Wide Usability**: Easy to use and very less technical difficulties making it usable for every stakeholders or individuals

Challenges

- **Limited Detail in LOD-1 Models**: May reduce accuracy for complex buildings
- **Shading Complexity**: Shading simulations can be less precise due to vegetations
- **Diverse User Needs**: Different stakeholders require varying levels of detail
- **High Computational Demand**: Real-time analysis for large cities could be resource-intensive

How to Overcome?

- **Hybrid Models**: Use higher-resolution data for key areas to enhance accuracy
- **Improved Shading Algorithms**: Option to input additional shading information
- **Customizable UI**: Offer adjustable detail levels for different users
- **Cloud Computing**: Use cloud-based processing to manage large data efficiently



Lack of accuracy

Shadow due to vegetation

IMPACTS AND BENEFITS

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Potential Impact on Target Audience

Policy-Makers:

- Informed decisions on urban renewable energy strategies.
- Promotes solar adoption in urban planning and sustainability policies.

Solar Solution Providers:

- Expands business opportunities with precise BIPV potential assessments.
- Improves accuracy of solar installations, optimizing energy output.

Architects & Urban Planners:

- Integration of solar energy solutions into building designs.
- Enhanced urban resilience through sustainable energy infrastructure.

Benefits of the Solution

Social:

- Promotes energy equity by making renewable energy more accessible in urban environments.
- Supports the creation of green jobs in solar energy sectors.

Economic:

- Reduces energy costs for consumers and businesses through optimized solar installations.
- Attracts investment in solar infrastructure development, boosting local economies.

Environmental:

- Reduces carbon footprint by enabling broader adoption of solar energy.
- Contributes to urban sustainability and the transition to clean energy sources.

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RESEARCH AND REFERENCES

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